

	MedTown Powder Dispenser Standard Operating Procedure	SOP # Pending Approval Effective Date: Pending Approval
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1.0 Standard Operating Procedure (SOP) Purpose

The purpose of this SOP is to provide clear instructions for maintaining Team MedTown's automated powder dispenser for the MedRock Lab Tech Internship. This document will outline how to disassemble/reassemble the device for dishwasher cleaning, recalibrate the scale, and perform other miscellaneous maintenance tasks. This SOP aims to give a clear understanding to any MedRock employee on how to maintain the device, regardless of their technical experience.

2.0 Procedures

2.1 Disassembly/Reassembly for Cleaning

Powder will naturally accumulate on the surfaces of the DOSER with time. These steps show how to safely disassemble the DOSER so that powder-contact parts are isolated for cleaning.

1. Remove power from the device by either unplugging it or using the ESTOP.
2. Pull the HOPPER forward (towards the user) until the set screws on the COUPLER are visible.
3. Use the 2mm ALLEN KEY from the BL extrusion to loosen the 2 set screws.
4. Pull the AUGER out of the COUPLER. The AUGER is now separate and can be cleaned.
5. Pull the HOPPER out of the EXTRUSION now that the AUGER is no longer in the way. The HOPPER is now separate and can be cleaned.
6. Complete the previous steps in reverse after cleaning to reassemble the DOSER after cleaning.

2.2 SCALE Recalibration

Since our software can correct the slight inaccuracies in our scale, recalibration is unnecessary.

2.3 Emergency Stopping

In the case of an emergency, the ESTOP will be pressed to cut off power from all of the electronics. After pressing, these steps should take place to keep both the device and employees safe.

1. Remove power completely from the device by unplugging it from the wall outlet. Do NOT rely on software commands to halt activity.
2. If an electrical fire occurs, then keep the base closed initially to limit the fire of oxygen. Keep a fire extinguisher ready in case the fire persists. Do not use the extinguisher unless the fire leaves the device, as water will permanently damage the electronics in the BASE.
3. When time elapses, power is fully gone, and the situation is safe to handle, remove all lids from the BASE if an electrical issue occurred. If the issue was mechanical, then clear what caused the

issue: this may include removing debris, manually fixing a jam in the DOSER, or tightening a structural screw with ALLEN KEYS.

4. Inspect the issue. If the issue was electrical, look for any visible damage to the wires or PCBs in the BASE. If the issue was mechanical, look for any damage done to the SERVO, COUPLER, or any other part.
5. Once all problems are inspected and fixed, plug the device back in and run test programs to investigate issues further. If issues are unresolved, then contact the SOP owner.

3.0 Definitions

For this SOP, the terms below have the following meanings:

Auger: The part attached to the COUPLER that is shaped like a screw and pushes the powder from the HOPPER to the SCALE.

Hopper: The part attached to the EXTRUSION that holds the powder and provides support for the AUGER.

Servo Mount: The part attached to the EXTRUSION that provides mounting for the SERVO.

Servo: The actuator mounted with the SERVO MOUNT and controlled by the SERVO DRIVER that actuates the COUPLER.

Coupler: The part that mechanically links the gear of the SERVO to the AUGER.

Extrusion: The parts attached to the BASE that provide mounting for the HOPPER and SERVO MOUNT.

Base: The core structure of the device, composed of 4 SHELLS and LIDS, which holds the major electronics (like the RPI, SCALE, SCREEN, etc.) and the EXTRUSION.

Shells: The bottom 4 parts of the BASE. Composed of the FL (front left), FR (front right), BL (back left), and BR (back right) shells.

Lids: The top 4 parts of the BASE. Composed of the FL (front left), FR (front right), BL (back left), and BR (back right) lids.

RPI: The central computing module of the device, located on the FL BASE. Controls SCREEN, SCALE, and SERVO DRIVER.

Power Supply: The central power device, located on the FL and BL BASE. Delivers power to the ESTOP and BUS BARS.

ESTOP: Large button located on the FL BASE that allows immediate shutoff of power from the POWER SUPPLY to the BUS BARS in case of emergency.

Bus Bars: The 2 rows of terminal posts for 12V power and ground, located on the FL and BL BASE. Distributes power to all components.

Servo Driver: The circuit board that takes serial input from the RPI and power from the BUS BARS and converts it into output that moves the SERVO, located on the BL BASE.

Scale: The device that weighs the powder, located on the FR and BR base. Uses a 9V BUCK CONVERTER for power and a TTL CONVERTER paired with a NULL MODEM for data.

Screen: The device that displays the interface, located on the FR base. Uses a 5V BUCK CONVERTER for power and connects via DSI to the RPI.

Buck Converter: The device that converts the 12V from the BUS BARS to either 5V (for the SCREEN, located on the FL BASE) or 9V (for the SCALE, located on the BR BASE)

TTL Converter: The device that converts the serial signal from the RPI from jumper wires to a DB9 connector that runs through the NULL MODEM, eventually connecting to the SCALE.

NULL MODEM: The device that converts the straight-through communication from the TTL CONVERTER to the null-modem communication that the SCALE uses.

ALLEN KEY: The tools used to loosen and tighten screws, located in the BR (2mm), BL (3mm), and FL (2.5mm) EXTRUSION holes.

DOSER: The structure of the device that controls powder dispensing, composed of the HOPPER, AUGER, SERVO, SERVO MOUNT, EXTRUSION, and COUPLER.

4.0 SOP Owner and Contact

4.1 SOP Owner: MedTown Team Lead

4.2 SOP Approved by: Grant Powell

4.3 Contact Information: MedTown Team Lead

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